Magnetism: electrostatics plus relativity

Nonrelativistic test charges around steady currents; c = 1/sqrt(mu0 eps0).

Two charges in motion

$$F_m = \frac{\mu_0}{4\pi}\frac{q_1 q_2 v_1 v_2}{r^2}$$

$$\frac{F_m}{F_e} = \frac{v_1 v_2}{c^2}$$

Neutral wires hide the electric force but not the magnetic one. Between two free charges moving parallel, the Biot-Savart field of one crossed with the velocity of the other gives F_m = (mu0/4pi) q1 q2 v1 v2 / r², while the Coulomb force is (1/4pi eps0) q1 q2 / r². Using mu0 eps0 = 1/c² the ratio is v1 v2 / c².

WHY / The tiny ratio is the reason magnetism looks like a separate force at laboratory speeds; it is the first relativistic correction to electrostatics.

WATCH / Only for v much less than c and parallel motion does this one-line ratio hold; perpendicular components involve gamma factors.

Charge density changes with frame

$$\lambda' = -\gamma I v/c^2, \qquad F = qvB$$

In the wire frame the positive and negative line densities cancel. A test charge moving at v sees length-contraction differences and a net force that, expanded to first order in v/c, equals q v B with B = mu0 I / 2 pi r. The constant mu0 = 1/(eps0 c²) emerges, not postulates.

WHY / Deriving magnetism from one boosted frame explains why the magnetic constant is tied to c: it is not an independent phenomenon.

WATCH / The full Lorentz transformation of fields is the proper language; the force-balance sketch is the exam-level derivation, not a replacement for Maxwell theory.

The worked comparison

$$F_e = ke^2/r^2 \gg F_m \text{ at lab speeds}$$

$$B = \frac{1}{c^2} v \times E$$

Two electrons at 10^6 m/s one centimetre apart: F_e about 2.3e-24 N and F_m smaller by 1e-11. In wires the electric force cancels to neutrality and the magnetic residue is what remains measurable. A moving charge field is B = v cross E over c² exactly in the nonrelativistic limit.

WHY / The estimator F_m/F_e = v1 v2/c² is a one-line validity check on any claimed magnetic dominance.

WATCH / For charges near light speed the magnetic force is comparable to the electric one; relativity is then not a correction but the description.

Limits of the classical picture

$$g = 2 \text{ (no classical account)}$$

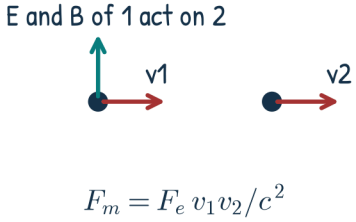
$$\mu_0 \varepsilon_0 = 1/c^2$$

Spin, the g factor, exchange coupling and superconductivity have no classical derivation; the relativistic picture covers orbital moments and forces between currents. Radiation from accelerating charges lies in the electromagnetic-waves module.

WHY / Stating the boundary of the classical model is part of the physics; the constant 1/c² is the bridge, not the whole road.

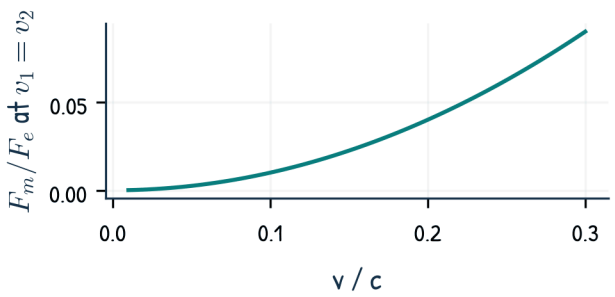
WATCH / Do not present v cross E / c² as valid at relativistic speed; transform the fields properly there.

Two moving charges: the magnetic term is the relativistic residue



$$F_m = F_e\, v_1 v_2/c^2$$

Magnetic force is smaller than electric by v squared over c squared



REMEMBER / The magnetic force between parallel moving charges is the electric force times v1 v2 / c²: magnetism is relativity with the constant coming out right.

The Helmholtz pair: engineering flatness

Two identical coaxial circular coils, equal currents, axial field.

The flatness condition

$$d = R:\ B(0) = \frac{8}{5\sqrt{5}}\frac{\mu_0 N I}{R}$$

$$B(z) = B_0\left[1 - \tfrac{144}{125}\left(\tfrac{z}{R}\right)^4 + \ \ldots\ \right]$$

Each coil contributes mu0 N I R² / 2(R²+z²)^{3/2} on axis at offset z from its centre. Expanding about the midpoint, odd terms cancel between coils and even ones add; the second derivative vanishes when the half-separation is R/2, i.e. spacing d = R. Then B(0) = 0.716 mu0 N I / R and the leading error is the fourth power of z/R.

WHY / The design rule d = R is a calculus result on the axial formula, not an empirical recipe.

WATCH / The field is flat on axis at the centre only to the stated order; radial uniformity and ends need their own analysis.

Worked design

$$N = 100,\ R = 20\ \text{cm},\ I = 1\ \text{A}:\ B = 0.45\ \text{mT}$$

Flat to 0.1 per cent within |z| < 0.17 R = 3.4 cm: a 7 cm uniform region. To cancel the Earth 45 microtesla inside the pair, drive 0.10 A with the axis along the local field. Coils in opposition can null the field entirely at the cost of the flat region.

WHY / Working one design end-to-end shows the trade between field size, uniformity and current.

WATCH / Uniform to 0.1 per cent is a specification; asking uniform without a tolerance invites wrong answers.

The second method

$$B''(0) = 0 \text{ at } s = R/2$$

inflection of one loop field

Expand each coil field separately: the sum of two identical coils at symmetric positions has no odd terms; the quadratic term carries B double-prime of one loop evaluated at half-separation. A single loop axial field has its inflection at s = R/2, exactly where the partner must sit. Coils in anti-Helmholtz (opposite currents) give a linear gradient instead.

WHY / Anti-Helmholtz is the trap question: opposite currents make a quadrupole used in atom trapping, not a flat field.

WATCH / Collapsing the coils to d = 0 gives one 200-turn loop: B = mu0 N I / R = 0.63 mT - a clean limit check of the 0.716 factor.

Numbers and checks

$$0.716 \to 1\ \ (d \to 0)$$

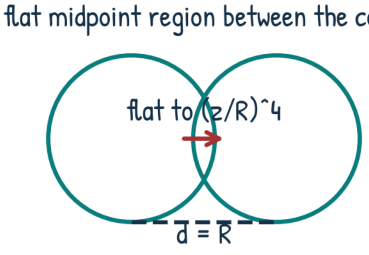
$\mu_0 NI/R$ is tesla

Dimensions of mu0 N I / R are tesla; d to zero returns the single-loop limit; large d splits into two independent loop fields. At the midpoint the field is uniform in the axial direction to fourth order - the entire point of the construction.

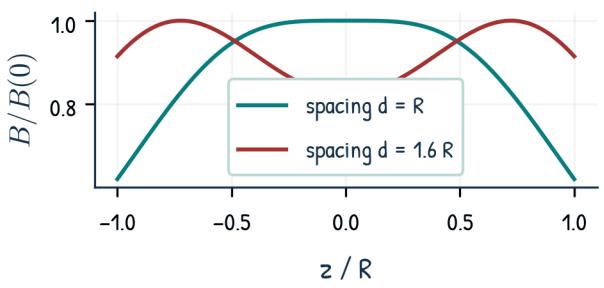
WHY / Limit checks verify the constant 8/(5 sqrt 5) = 0.7155 before trusting uniformity claims.

WATCH / Helmholtz uniformity is not spherical uniformity; a large sample sees radial falloff even at the magnetic centre.

Helmholtz spacing d = R cancels the quadratic term at the centre



Only d = R flattens the midpoint; wider spacing gives a dip



REMEMBER / Spacing the coils one radius apart cancels the second derivative at the midpoint; uniformity is then fourth order.

The magnetised sphere two ways

Uniform M in a sphere; demagnetising factors; external dipole field.

Rings versus demagnetising factor

$$B_{\text{in}} = \tfrac{2}{3}\mu_0 M$$

$$\mathbf{H} = -N_d \mathbf{M}, \quad N_d = \tfrac{1}{3}$$

Method 1 stacks the bound surface current K = M sin theta as rings and integrates the axial field at the centre. Method 2 uses the general result that inside a uniformly magnetised ellipsoid the demagnetising field is -N_d M with N_x + N_y + N_z = 1; a sphere shares equally, N = 1/3, giving B = mu0(M - M/3). The second method proves the interior is uniform, not just the centre.

WHY / Two methods answer different weaknesses: the ring integral gives one point, the demagnetising argument gives the whole interior.

WATCH / N_d = 1/3 is a sphere result; a long cylinder has N = 0 and a thin plate N = 1. Do not reuse one third for other shapes.

Worked numbers

$$M = 10^6 \text{ A/m}, \quad R = 1 \text{ cm}$$

$$\mu = 4.19 \text{ A m}^2$$

Interior B = 2/3 mu0 M = 0.84 T. Moment is M times volume: 4 pi R cubed over 3 times M = 4.19 A m². At r = 5R on the axis the dipole field is 6.7e-3 T. A long cylinder of the same M would have 1.26 T inside - larger because its demagnetising field vanishes.

WHY / Comparing shapes at fixed M isolates geometry as the variable; the moment scales with volume in every case.

WATCH / The far-field formula needs r much larger than R; at r = 5R the dipole approximation is already per cent level at the pole.

Matching the rotating shell

$$K = M \sin\theta$$

$$B_{\text{pole}} = \tfrac{2}{3}\mu_0 M \text{ continuous}$$

The surface current of a uniformly magnetised sphere equals that of a rotating charged shell with sigma omega R = M. Just outside the pole the dipole field is also 2/3 mu0 M, continuous with the inside as the normal component must be; at the equator outside it is -1/3 mu0 M, a tangential jump mu0 K as required.

WHY / Continuity checks on B_normal and the K jump on tangential H are the field-matching exam skill.

WATCH / Inside the sphere the field is uniform yet the surface current varies with theta; do not expect a solenoid profile.

Sensitivity to shape

$$N_x + N_y + N_z = 1$$

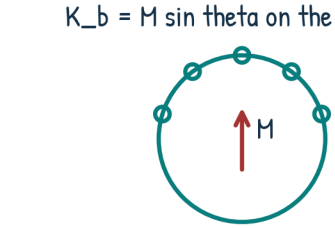
$$N_{\text{sphere}} = \tfrac{1}{3}$$

Ellipsoids are the only shapes with uniform interior field; anything else has non-uniform M response. Laboratory magnetisation measurements therefore quote the sample shape with chi; a powder and a single crystal of the same material can differ.

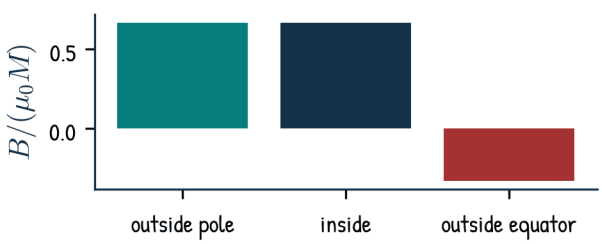
WHY / Demagnetising corrections are the first systematic error in any susceptibility measurement.

WATCH / Reported chi without shape is ambiguous; ask for the demagnetising factor before believing the number.

Uniformly magnetised sphere: surface current of a rotating shell



Sphere field: uniform inside, dipole outside with a tangential jump



REMEMBER / Interior field two-thirds mu0 M; outside a pure dipole; the demagnetising factor of a sphere is one third.

Magnetic pressure at work

Surface currents confining field; plasma channels; levitated slabs; cosine-theta windings.

Solenoid and pinch

$$P = \frac{B^2}{2\mu_0}$$

$$B_{\text{surface}} = \frac{\mu_0 I}{2\pi a}$$

A 3 T solenoid of 10 cm² bore: 3.6 MPa hoop stress (36 atm) and 3.6 kN pulling the halves together. A 30 kA lightning stroke in a 5 mm channel makes 1.2 T at its surface and pinches itself with 0.57 MPa of inward magnetic pressure, balanced by plasma thermal pressure (the Bennett relation).

WHY / The same formula is outward on a solenoid winding and inward on a plasma column: pressure acts on whatever confines the field.

WATCH / The pinch is unstable (sausage and kink modes); quoting the equilibrium pressure is not a stability proof.

Levitated slab and numbers

$$B = \sqrt{2\mu_0 mg/A}$$

$$1\ \text{T} \rightarrow 4\ \text{atm}$$

A type-I superconductor excludes B entirely and feels the full pressure from outside: supporting 2.0 kg over 4.0 cm² needs 0.35 T at the plate. The Meissner effect makes levitation a stress problem; type-II pinning adds hysteresis and lateral stability.

WHY / Pressure methods give one-line levitation estimates that match image-dipole calculations to a factor of two.

WATCH / A field of 1 T supports a steel slab 5 m thick in principle; practical limits are the superconductor, not the stress.

Cosine-theta coil

$$K = K_0 \cos\varphi$$

$$B_{\text{inside}} = \frac{\mu_0 K_0}{2}$$

Let two overlapping current cylinders of the overlap trick approach: their net surface current is K0 cos phi around the cylinder and the interior field stays uniform and transverse. Every accelerator dipole magnet is this construction in superconducting cable, held by steel collars against 27 MPa at 8.3 T.

WHY / The overlap trick taken to its limit designs the LHC dipole: same field theory, engineering scale.

WATCH / Uniform interior requires the full cosine distribution; approximating it with discrete coils leaks multipole errors that steer beams.

Stress checks in numbers

$$P = \frac{B^2}{2\mu_0}$$

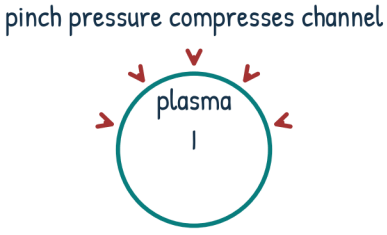
$$1\ \text{T} \rightarrow 3.98 \times 10^5\ \text{Pa}$$

At 3 T the hoop stress is 36 atmospheres; at 8.3 T it is 27 MPa, the LHC collar design point. Remember 1 atm = 1e5 Pa so B squared over 2 mu0 at 1 T is about 4 atmospheres: a two-line sanity check on any magnet mechanical question.

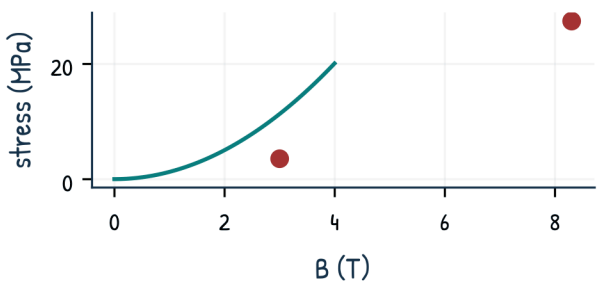
WHY / A pressure-of-field table converts stress answers instantly and catches unit slips in Pa versus atm.

WATCH / Stress acts on the current distribution, not on the vacuum; quoting B squared over 2 mu0 as a pressure in empty space needs the confining current named.

Current channel pinched by its own magnetic pressure



Hoop stress from B squared: 3.6 MPa at 3 T, 27 MPa at 8.3 T



REMEMBER / B²/2 mu0 is engineering: hoop stress, pinch, levitation and accelerator dipoles all obey it.

The cycloid family and Fermi acceleration

Crossed uniform fields; then slowly moving magnetic mirrors.

Classify by drifting frame

$$v_d = E/B, \quad r = \frac{m|\mathbf{v}_0 - \mathbf{v}_d|}{qB}$$

$$\text{wheel radius } r_c = \frac{mE}{qB^2}$$

In the drifting frame the particle circles at speed |v0 - vd|; the lab path is that circle carried along at vd. v0 = vd is a straight selector line; |v0 - vd| = vd is a cusped cycloid; larger gives loops (prolate), smaller curtate arches. Protons in E = 2e3 V/m, B = 0.020 T: vd = 1e5 m/s, r_c = 5.2 cm.

WHY / The drifting-frame construction classifies every initial condition in one picture; solving lab equations of motion reproduces it.

WATCH / Cusped, curtate and prolate refer to the gyration radius relative to the wheel, not to different forces.

Energy across one loop

$$v_{\text{top}} = 3v_d \text{ in a loop}$$

$$\Delta K = qE \cdot 2r$$

Energy conservation over one arch gives the top speed from qE times height 2r; in a prolate loop the bottom speed exceeds 3vd and the top falls to vd, checked by half m v² differences. The drift is the same for an electron; only the circle size changes by the mass ratio.

WHY / Energy accounting verifies trajectory classification without integrating equations of motion.

WATCH / The proton and electron arch heights differ by 1836 even in identical fields; never transfer one orbit dimension to the other.

Fermi acceleration

$$\frac{\Delta K}{K} \approx \frac{4V}{v}$$

$$N \approx 0.21\frac{v}{V} \text{ to double}$$

A particle reflecting from an approaching mirror gains 2V per head-on bounce in speed, hence about 4V/v in energy. Doubling energy takes about 0.21 v/V reflections: for cosmic rays on 10 km/s clouds that is 6000 bounces over 6000 years per light-year, and receding mirrors cancel half the gain at second order - shock acceleration is the modern resolution.

WHY / First-order (head-on shocks) and second-order (random clouds) Fermi acceleration differ in efficiency, not in principle.

WATCH / The adiabatic invariant p_parallel times path length gives the same gain law for slow compression; do not mix adiabatic and impulsive regimes in one estimate.

Cycloid numbers worked

$$E = 2.0 \times 10^3 \text{ V/m}, \quad B = 0.020 \text{ T}$$

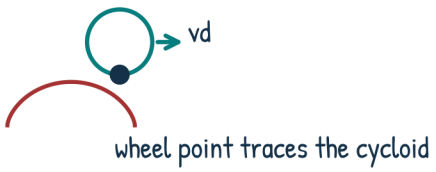
$$v_d = 1.0 \times 10^5 \text{ m/s}$$

Protons: wheel radius r_c = m v_d / eB = 5.2 cm and gyro period 3.3 microseconds. Launch along +x at vd gives a straight line; at 2 vd a cusped cycloid; from rest the classic arch of height 10.4 cm; at minus vd prolate loops with instantaneous reverse motion.

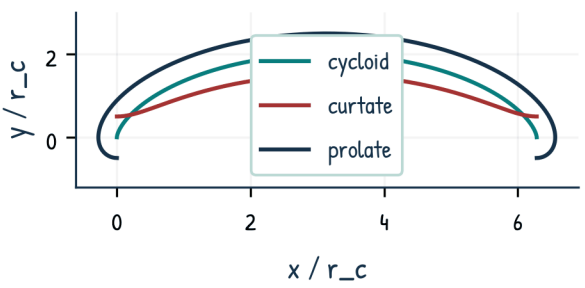
WHY / Four initial conditions, one construction: classify with |v0 - vd| before integrating anything.

WATCH / In a prolate loop the instantaneous velocity reverses at the bottom of each loop; do not assume monotone forward motion.

Rolling wheel picture: the gyration centre drifts at E cross B over B squared



Wheel-point distance sets cycloid family: curtate, cusped or looping



REMEMBER / Every crossed-field orbit is a rolling wheel; every reflection off an approaching mirror gains energy.

Three clocks and a synchrotron

Trapped particles in dipole fields; ultrarelativistic electrons in ramps.

Gyration, bounce, drift

$$T_c = \frac{2\pi m}{eB}$$

$$T_d \approx \frac{2\pi e B r^2}{3K}$$

A 1 MeV proton at L = 2 in the Earth field (3.9 microtesla): gyration 17 ms with a 37 km radius, bounce between mirrors about 2 s, gradient drift around the Earth about 1300 s westward. The three timescales separate by two orders each, making each motion an adiabatic invariant of the next.

WHY / Hierarchy of clocks is what makes guiding-centre theory work; check the separation before applying invariants.

WATCH / Curvature drift adds to gradient drift for particles with parallel motion; quote which drift the period includes.

Drift period scaling

$$T_d \propto \frac{1}{K}$$

$$\text{ring current of MA}$$

Higher-energy particles drift faster and circle sooner: a 10 MeV proton takes 2 minutes, a 10 keV one 1.5 days. The trapped population carries a westward ring current of a few mega-amperes, the geomagnetic storm signature.

WHY / Energy-dependent drift times explain why injections of energetic particles disturb compasses within minutes.

WATCH / Dipole field model and local time effects give order-of-magnitude periods; exact drift shells need full field models.

The synchrotron

$$f = \frac{eB}{2\pi\gamma m}$$

$$\Delta E_{\text{turn}} = 88.5\ \text{keV}\frac{E^4[\text{GeV}]}{r[\text{m}]}$$

A 1 GeV electron in 1 T has orbit radius 3.3 m and revolution 14.3 MHz; the cyclotron formula would say 28 GHz - wrong by gamma = 1958. The synchrotron holds r fixed by ramping B with momentum and tunes the cavity towards c/2 pi r. Electrons radiate 27 keV per turn at that energy and must be replenished.

WHY / Synchrotron radiation sets the energy ceiling of electron machines and is itself a light source.

WATCH / Radiation loss scales as the fourth power of energy: small energy increases, large losses. Classical cyclotron answers for electrons are always wrong.

Detector-scale numbers

$$p[\text{GeV}/c] = 0.3\,B[\text{T}]\,r[\text{m}]$$

$$E_e \text{ radiates, } E_p \text{ less}$$

The 300 Br rule in GeV units is the workhorse of collider design. Proton machines radiate far less at the same energy because the loss scales as 1/m⁴ at fixed energy; this is why the LHC collides protons while light sources exploit electron radiation.

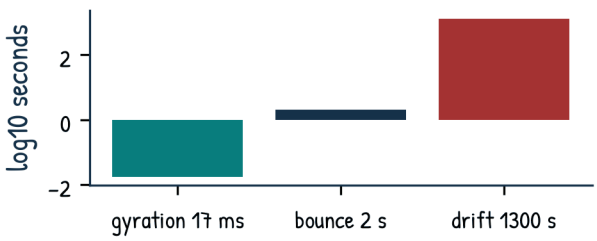
WHY / Knowing which particle radiates decides accelerator architecture and detector backgrounds.

WATCH / At fixed radius and B, momentum is fixed but energy depends on mass; always state which quantity the machine holds fixed.

One trapped particle keeps three clocks: gyration, bounce, drift



Three clocks separated by two orders each make adiabatic invariants



REMEMBER / Radiation-belt particles keep three well-separated clocks; a synchrotron abandons the fixed cyclotron frequency.

Two carriers, levitation and why iron beats water

Mixed carriers in semiconductors; image dipoles over perfect diamagnets; susceptibility estimates.

Two-carrier Hall

$$R_H = \frac{p\mu_p^2 - n\mu_n^2}{e(p\mu_p + n\mu_n)^2}$$

With electrons and holes both mobile, the transverse current balance weights each by the square of its mobility: R_H vanishes when n mu_n² = p mu_p² and flips sign beyond. Germanium at room temperature is p-type by count yet Hall-positive; heated to n = p it becomes Hall-negative, R_H = -7.2e-3 m³/C.

WHY / Mobility weighting, not carrier counting, decides the Hall sign: fast electrons dominate slow holes of equal number.

WATCH / The vanishing line n/p = (mu_p/mu_n)² is where R_H passes through zero; near it the simple one-carrier formula is doubly wrong.

Levitation by images

$$h = \left(\frac{3\mu_0\mu^2}{32\pi mg}\right)^{1/4}$$

$$F = \frac{3\mu_0\mu^2}{2\pi s^4}$$

A perfect diamagnet reflects a dipole into an antiparallel image at the mirror depth; coaxial antiparallel dipoles at separation 2h repel as 3 mu0 mu² / 32 pi h⁴. A 1 cm³ NdFeB cube of moment 1 A m² floats 2.7 cm above a superconductor; doubling its weight only lowers h to 2.3 cm.

WHY / The quarter-power makes levitation forgiving: masses and moments change heights slowly.

WATCH / Type-II pinning adds lateral stability that the ideal image calculation cannot give; ideal Meissner levitation is laterally unstable.

Iron versus water numbers

$$\chi_{\text{water}} \approx -9 \times 10^{-6}$$

$$k_B T_c \approx zJ/3$$

Dipole coupling between atomic moments is 0.04 K worth of energy - nothing at room temperature. Exchange J about 0.05 eV with z = 8 neighbours gives Tc near 1500 K, the right order for iron 1043 K. Water diamagnetism from the Larmor formula with Z = 10 and <r²> = (0.07 nm)² gives -9.5e-6 against the measured -9.0e-6.

WHY / Three order-of-magnitude estimates in one problem: rule out the wrong mechanism, estimate the right one, verify the weak universal effect.

WATCH / Classical Larmor diamagnetism gets the right scale only with a sensible <r²>; the agreement for water is partly fortunate in that choice.

Perfect versus Larmor diamagnet

$$\chi = -1 \ \text{(Meissner)}$$

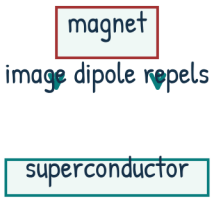
$$\chi_{\text{Larmor}} \sim -10^{-5}$$

A superconductor screens B completely with surface currents and floats magnets centimetres high; ordinary diamagnets reduce the field by parts per million and levitate nothing visible. Both are negative susceptibilities; only one reaches the Meissner limit. Levitation estimates use image dipoles or B squared over 2 mu0 stress.

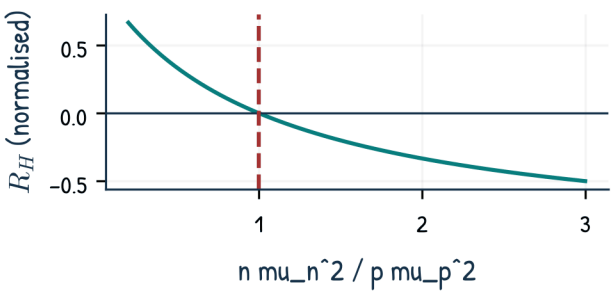
WHY / Separating the two mechanisms prevents the classic error of using chi = -1e-5 in a levitation estimate.

WATCH / Type-II superconductors trap flux and pin vortices: ideal Meissner image formulas describe type-I or reversible type-II only.

Levitation: a perfect diamagnet reflects the dipole into an opposite image



Two-carrier Hall sign flips when n mu_n squared overtakes p mu_p squared



REMEMBER / Mixed carriers can invert the Hall sign; levitation heights follow from image forces; iron orders by exchange.

Estimates, Gauss method and model limits

Order-unity geometry factors; dipole model of the Earth; classical magnetostatics assumptions.

Gauss absolute measurement

$$\mu B_H = \frac{4\pi^2 I_m}{T^2}, \qquad \frac{\mu}{B_H} = \frac{\tan\theta\, d^3}{2\times 10^{-7}}$$

Oscillating a magnet of moment of inertia I_m gives the product mu B_H; deflecting a compass from distance d gives the ratio mu / B_H. I_m = 6e-6 kg m², T = 4.0 s, d = 20 cm, theta = 30 degrees give mu = 0.58 A m² and B_H = 25 microtesla: a field in mechanical units at last.

WHY / Two measurements, two unknowns: the structure of the method is the physics, not the arithmetic.

WATCH / The deflection formula needs the magnet far enough for its dipole field and the compass in its meridian; both are standard idealisations.

Estimates that teach

$$I_{\text{core}} \sim 2\times 10^{9}\ \text{A}$$

$$B_{\text{heart}} \sim 100\ \text{pT}$$

A core-sized current loop needs two billion amperes, hence a dynamo. A compass within 10 cm of a 0.25 A wire errs a degree; a car battery lead must sit a metre away. The heart current dipole of 1e-5 A m produces 100 pT at 10 cm, measurable only with SQUIDs near the flux-quantum noise floor.

WHY / Each estimate answers a question a curious examiner can ask; the physics is the scaling, the number is the check.

WATCH / Bioelectric current dipoles are ampere-metres with 1/r² fields; orbital magnetic dipoles are ampere-metre-squared with 1/r³ fields. Mixing the two powers is the classic slip.

Where the model breaks

$$\gamma = qB \text{ failures: } f = \frac{eB}{2\pi\gamma m}$$

$$\chi = -1 \ \text{(superconductor)}$$

Relativity moves the cyclotron frequency by 1/gamma; circling charges radiate; Landau quantisation and g = 2 are quantum; iron saturates at 2.1 T and its mu_r is a loop slope; changing flux induces electric fields - the next module; plasmas add collective effects and instabilities.

WHY / Listing failure modes is the mature end of a physics chapter: knowing when not to trust the formulas.

WATCH / Superconductivity and exchange are quantum from the ground up; classical analogies there are metaphors, not derivations.

Consistency identities

$$\tan\theta_{\text{dip}} = 2\tan\lambda$$

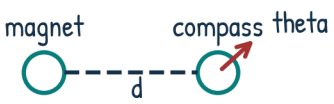
$$B = B_{\text{eq}}\sqrt{1+3\sin^2\lambda}$$

A station with dip 60 degrees sits at magnetic latitude 41 degrees in the dipole model; its total field is 1.5 B_eq. Checking both identities on any station table separates model agreement from local anomaly data. The dipole moment from B_eq = 30 microtesla gives 7.8e22 A m².

WHY / The dip-latitude identity is a free consistency check in every Earth-field question.

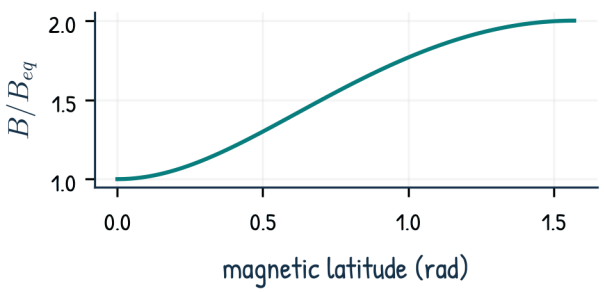
WATCH / Dipole geometry assumes a centred, aligned dipole; the 11 degree tilt and secular drift spoil precision but not the estimates.

Absolute measurement: magnet oscillation plus compass deflection at distance d



oscillation: product; deflection: ratio

Dipole field strength grows from equator to pole by up to a factor two



REMEMBER / Estimates are arguments: each one shows what the model can and cannot do.